

Product Requirements Document (PRD)

Project Name: FixEdu: School Facility Condition Reporting & Repair Tracking Portal

1. Objective & Goals

The goal of FixEdu is to streamline facilities maintenance. The system enables students and teachers to submit reports of damaged facilities, which are then assigned to maintenance staff by administrators. Staff can start and mark these repairs complete with before/after proof images.

2. User Roles

- **Students & Teachers (Reporters):** Submit issues, upload proof images, and trace statuses on interactive vertical timelines.
- **Maintenance Staff:** Track assigned jobs, change status to In Progress, and submit completion images/notes.
- **Administrators:** Full oversight dashboard, assign tickets, manage facilities registry (CRUD), inspect user lists, view analytics charts.

3. Technical Features

- JWT auth with bcrypt.
- File upload limits (up to 5 images, 5MB).
- Mongoose/MongoDB Atlas database setups.
- Query filters, pagination parameters, and regex text searches.
- Automated Nodemailer email dispatches on reports, assignments, and completions.

4. Technical Stack

- React.js (Vite) / React Router v6 / Chart.js / Axios

- Node.js / Express.js / Mongoose
- MongoDB Atlas Cloud
- Nodemailer / Multer / Helmet / express-rate-limit / express-mongo-sanitize